US Brewery Market Analysis

Finding the Best State to Open a New Brewery

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Intro & Data | EDA | Model & Results | Conclusion

Data Sources & Pipeline

Target Audience: Brewery entrepreneurs and investors seeking data-driven location selection

API	API	CSV
Open Brewery DB	**US Census ACS 5-Year**	**Zillow ZORI**
8,126 US breweries Name, type, city, state, coordinates	Population & median household income by city & state (2020–2024 avg)	State-level average rent 50 states covered (2020–2024 avg)

API Pull → Quality Check & Clean → Feature Eng. → SQLite Load & SQL Analysis → EDA → Regression & Ranking

SQLite Integration: 3 tables (brewery_agg, census_agg, zillow_rent) → merged via JOIN → state_metrics → window functions for regional benchmarking

Key Metrics & Feature Engineering

Raw counts cannot be compared across states — standardization is essential for fair market assessment

Brewery Density

breweries / pop × 100k

→ *Competition level*

People / Brewery

population / breweries

→ *Market potential*

Income / Brewery

income / breweries

→ *Purchasing power*

Region (OHE)

5 regions one-hot encoded

→ *Geographic controls*

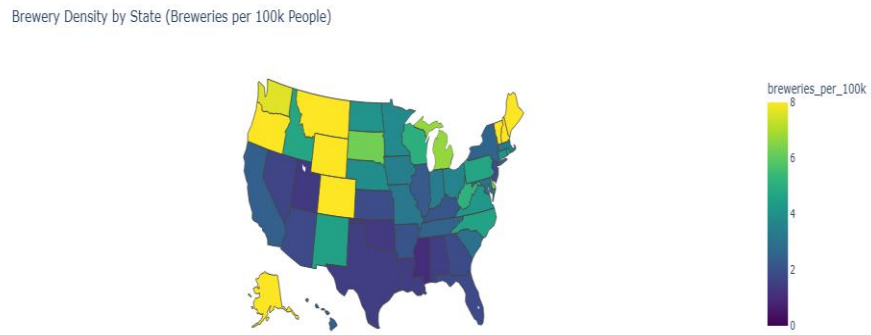
Transformations: Log(1+x) applied to reduce right skew → MinMax scaling [0,1] to unify scale for regression & composite ranking

Data Quality Notes

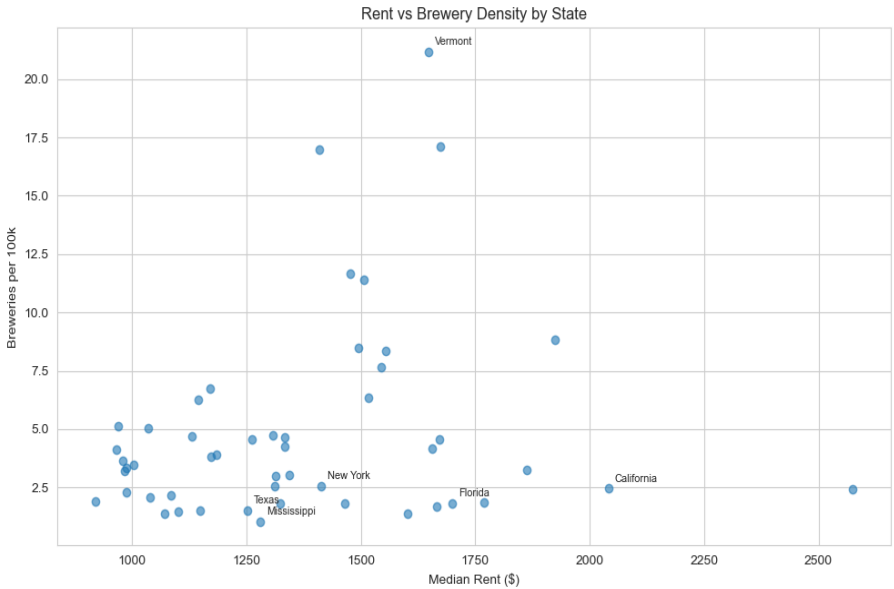
- Coordinate outliers (positive longitude / extreme latitude) → excluded from map only, rows retained
- Census API negative income values (−66,666,666 flag) → replaced with NA
- 994 city-level records unmatched after merge → analysis focused at state level (100% complete)

EDA: Competition Landscape & Rent Analysis

BREWERY DENSITY BY STATE



RENT vs. BREWERY DENSITY BY STATE



SW 2.31

Most underserved

3.7×
gap

NE 8.49

Most saturated

Rent ≠ density: No strong correlation — cost alone should not drive location decisions

Culture matters: Vermont leads at 21/100k despite $1,600 rent — policy & culture drive density

EDA: Opportunity States

13 Opportunity States Identified

Selection criteria: Above-average weighted income AND below-average brewery density per 100k — high purchasing power with low competitive saturation

	state	region	breweries_per_100k	weighted_avg_income	mean_rent	people_per_brewery
0	Maryland	Southeast	3.0256	109285.05	1344.45	33051.0
1	Virginia	Southeast	4.2406	106075.46	1333.03	23582.0
2	New Jersey	Northeast	1.8767	104574.84	1768.99	53284.0
3	California	West	2.4498	104267.42	2040.86	40820.0
4	Hawaii	West	2.4131	103139.48	2575.27	41440.0
5	Massachusetts	Northeast	3.2595	102647.78	1861.74	30679.0
6	Utah	West	1.3801	98563.18	1601.92	72460.0
7	Minnesota	Midwest	3.8095	92692.42	1172.32	26250.0
8	New York	Northeast	2.5681	90920.96	1414.25	38939.0
9	Connecticut	Northeast	4.1738	89587.88	1656.26	23959.0
10	Illinois	Midwest	2.2870	88164.92	987.56	43725.0
11	Arizona	Southwest	1.8114	83511.73	1465.82	55205.0
12	Texas	Southwest	1.5121	79958.83	1252.18	66134.0

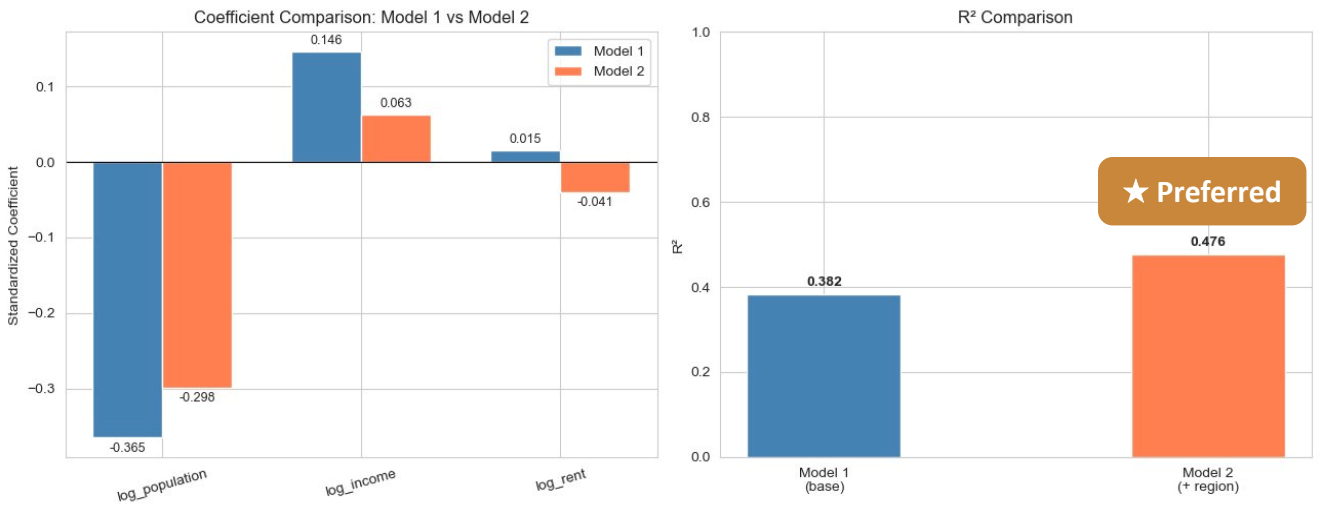
Sorted by income — Maryland and Virginia look great on paper, but they're already more competitive. Income alone doesn't tell the full story. To weight all these factors together, that's where regression comes in.

Regression Models

Model 1 $\log(\text{breweries_per_100k}) = \beta_0 + \beta_1 \log(\text{population}) + \beta_2 \log(\text{income}) + \beta_3 \log(\text{rent}) + \epsilon$

Model 2 $\log(\text{breweries_per_100k}) = \beta_0 + \beta_1 \log(\text{population}) + \beta_2 \log(\text{income}) + \beta_3 \log(\text{rent}) + \beta_4$ **region dummies** $+ \epsilon$

All features log-transformed; continuous predictors standardized (StandardScaler); in-sample metrics reported (n=50, train/test split not applicable)



	Model 1	Model 2 ★
R^2	0.382	**0.476**
MAE	0.358	**0.334**
RMSE	0.445	**0.410**

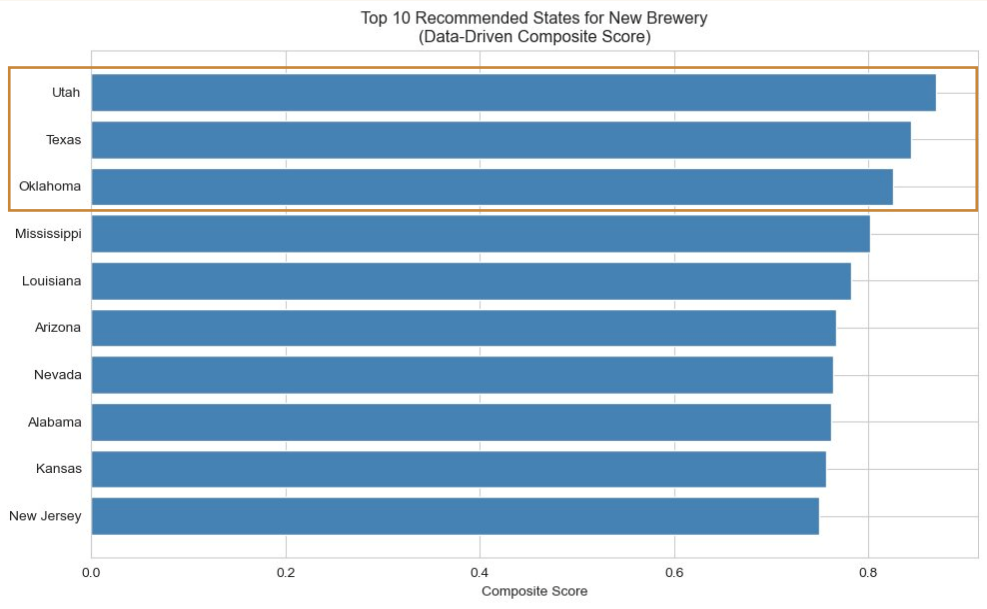
Model 2 (+region controls) preferred:

- R^2 increases from 0.382 to 0.476 — regional culture/geography explain an additional 9.4% of variation beyond economic factors alone
- MAE drops from 0.358 to 0.334, RMSE drops from 0.445 to 0.410 — region controls improves the model's predictions
- Rent flips negative — high-rent states were confounded with high income in Model 1; region controls isolate the true cost effect
- Coefficients are used as composite score weights — account for regional heterogeneity

Scoring, Recommendation & Conclusions

Weights from Model 2 coefficients (absolute value, normalized):

$$Score = 74.24\% \times \underbrace{rank(people_per_brewery)}_{\text{Underserved Market Size}} + 15.64\% \times \underbrace{rank(income)}_{\text{Purchasing Power}} + 10.12\% \times [1 - \underbrace{rank(rent)}_{\text{Cost of Entry}}]$$



Recommendation

large underserved markets, strong income, affordable rent

#1	#2	#3
Utah	**Texas**	**Oklahoma**
Score: 0.869	Score: 0.844	Score: 0.825

Limitations

- n = 50 states limits regression power
  — city-level analysis would improve results
- Zillow rent = residential, not commercial
  — may underestimate actual lease costs